Task-Background Document: From the Free-Fermion Action to the G–Σ and Coadjoint-Orbit Bosonization Actions

This document fixes the starting point and the two intermediate targets used throughout the project. Section 1 derives the bilocal (G–Σ) action of free, gapless, nonrelativistic fermions from the microscopic coherent-state action, and identifies its saddle point. Section 2 derives the coadjoint-orbit bosonization action from the same microscopic action, following the Appendix of Ye and Wang, *Phys. Rev. B* **112**, 035113 (2025). The conventions in §1.1 are common to both.

1 From the free-fermion action to the G–Σ action

1.1 Preliminaries and conventions

We work with spinless nonrelativistic fermions in d spatial dimensions at inverse temperature β, in the imaginary-time coherent-state path integral. The dynamical variables are Grassmann fields $\psi(\boldsymbol{x},\tau)$, $\bar{\psi}(\boldsymbol{x},\tau)$ that are antiperiodic in imaginary time, $\psi(\boldsymbol{x},\tau+\beta) = -\psi(\boldsymbol{x},\tau)$.

To keep the bilocal structure visible we use a collective coordinate $1 \equiv (\boldsymbol{x}_1, \tau_1)$, with

$$\int \mathrm{d}1 \;\equiv\; \int_0^\beta \mathrm{d}\tau_1 \int \mathrm{d}^d x_1, \qquad \delta(1,2) \;\equiv\; \delta(\tau_1-\tau_2)\,\delta^d(\boldsymbol{x}_1-\boldsymbol{x}_2). \tag{1}$$

The single-particle Hamiltonian is $\hat{h} = \varepsilon(-i\boldsymbol{\nabla}) - \mu$, with band dispersion $\varepsilon(\boldsymbol{k})$, chemical potential μ, and $\xi_{\boldsymbol{k}} \coloneqq \varepsilon(\boldsymbol{k}) - \mu$. "Gapless" means the Fermi surface $\{\boldsymbol{k} : \xi_{\boldsymbol{k}} = 0\}$ is nonempty and codimension one. The microscopic action is

$$S_0[\bar{\psi},\psi] = \int \mathrm{d}1\; \bar{\psi}(1)\left(\partial_{\tau_1} + \hat{h}_1\right)\psi(1). \tag{2}$$

It is convenient to write (2) as a bilinear $S_0 = -\bar{\psi}\, G_0^{-1}\, \psi$ in the free inverse propagator, defined as the operator with kernel

$$G_0^{-1}(1,2) \;\coloneqq\; -\left(\partial_{\tau_1} + \hat{h}_1\right)\delta(1,2). \tag{3}$$

The sign in (3) is chosen so that the Matsubara propagator takes its standard form: with $\partial_\tau \mapsto -i\omega_n$ on antiperiodic frequencies $\omega_n = (2n+1)\pi/\beta$,

$$G_0(\boldsymbol{k}, i\omega_n) = \frac{1}{i\omega_n - \xi_{\boldsymbol{k}}}. \tag{4}$$

We define traces of bilocal kernels by $\mathrm{Tr}(AB) := \int \mathrm{d}1\,\mathrm{d}2\, A(1,2)\, B(2,1)$, and $\mathrm{Tr}\ln$ through its series. Throughout, G_0 denotes the free *propagator* (4) and G_0^{-1} the operator (3); note $G_0^{-1} = -(\partial_\tau + \hat{h})$ carries an explicit minus sign relative to $\partial_\tau + \hat{h}$.

1.2 The bilocal field, and why to introduce it

The low-energy excitations of a Fermi sea are neutral particle–hole pairs, not single fermions. Their natural carrier is the equal-time, and more generally the bilocal, two-point function $\bar\psi(2)\psi(1)$: smooth deformations of the occupied region of phase space are encoded in its fluctuations. We therefore promote the bilocal to an independent field rather than treating it as a composite of $\psi, \bar\psi$.

Concretely, introduce a bosonic bilocal field $G(1,2)$ together with the constraint that it equal the microscopic bilinear,

$$G(1,2) \;=\; -\,\psi(1)\,\bar\psi(2), \tag{5}$$

where the minus sign is dictated by (4): at the level of expectation values (5) gives $\langle G(1,2)\rangle = -\langle \psi(1)\bar\psi(2)\rangle = -\langle T_\tau\, \psi(1)\bar\psi(2)\rangle = G_0(1,2)$, the conventional Green's function. For an *interacting* system this step is forced: a quartic interaction is decoupled by a field conjugate to a fermion bilinear, and $G(1,2)$ is the most general such collective field (it carries every particle–hole channel, not only the density). For the free system there is nothing to decouple, but the move still buys the essential thing — it trades the Grassmann fields for a bosonic bilocal whose path integral is *exact* and whose fluctuation spectrum is precisely the particle–hole continuum. That is the arena §2 reorganizes into geometric variables.

1.3 The self-energy as a Lagrange multiplier

We enforce (5) inside the path integral by a functional delta-function, represented through a second bilocal field $\Sigma(1,2)$ — the self-energy — integrated along the imaginary direction:

$$1 = \int \mathcal{D}G\,\mathcal{D}\Sigma\; \exp\Big(-\int \mathrm{d}1\,\mathrm{d}2\; \Sigma(2,1)\big[\,G(1,2) + \psi(1)\bar\psi(2)\,\big]\Big). \tag{6}$$

The Σ-integral produces $\prod_{1,2} \delta\big(G(1,2) + \psi(1)\bar\psi(2)\big)$, which collapses to the constraint (5); Σ is the field conjugate to the bilocal, and will turn out to be the self-energy. Inserting (6) into $Z = \int \mathcal{D}\bar\psi\,\mathcal{D}\psi\, e^{-S_0}$ leaves the action quadratic in the fermions.

1.4 Integrating out the fermions

Collecting the fermion-bilinear terms from $-S_0$ and from (6), and using $\psi(1)\bar\psi(2) = -\bar\psi(2)\psi(1)$,

$$-S_0 - \int \Sigma(2,1)\,\psi(1)\bar\psi(2) = \bar\psi\, G_0^{-1}\,\psi + \bar\psi\,\Sigma\,\psi = \bar\psi\,\big(G_0^{-1} + \Sigma\big)\,\psi, \tag{7}$$

with Σ read as the operator of kernel $\Sigma(2,1)$. The Grassmann Gaussian integral $\int \mathcal{D}\bar\psi\,\mathcal{D}\psi\, \exp[\bar\psi(G_0^{-1}+\Sigma)\psi] = \det\big(G_0^{-1} + \Sigma\big) = e^{\mathrm{Tr}\ln(G_0^{-1}+\Sigma)}$ (an additive, G,Σ-independent constant is dropped) leaves a purely bosonic theory,

$$Z = \int \mathcal{D}G\,\mathcal{D}\Sigma\; \exp\big(\,\mathrm{Tr}\ln(G_0^{-1} + \Sigma) - \mathrm{Tr}(\Sigma G)\big), \qquad \mathrm{Tr}(\Sigma G) := \int \mathrm{d}1\,\mathrm{d}2\; \Sigma(2,1)\, G(1,2). \tag{8}$$

1.5 The G–Σ action and its saddle point

We have arrived at the exact bilocal action

$$\boxed{S[G,\Sigma] = -\operatorname{Tr}\ln\bigl(G_0^{-1}+\Sigma\bigr) + \operatorname{Tr}(\Sigma G) + S_{\rm int}[G]} \tag{9}$$

where we have restored $S_{\rm int}[G]$, the interaction rewritten in terms of the bilocal (for a density–density interaction $\frac{1}{2}\int V\,\rho\rho$ this is a local functional of G); for the free fermions of interest $S_{\rm int}=0$. Equation (9) is the central object of this section: it is a *bosonic* action, obtained without approximation, whose two fields are the propagator G and the self-energy Σ.

The saddle-point equations follow from $\delta S/\delta\Sigma = 0$ and $\delta S/\delta G = 0$. Using $\delta\ \operatorname{Tr}\ln(G_0^{-1}+\Sigma) = \operatorname{Tr}[(G_0^{-1}+\Sigma)^{-1}\delta\Sigma]$,

$$\frac{\delta S}{\delta\Sigma(2,1)} = -\,(G_0^{-1}+\Sigma)^{-1}(1,2) + G(1,2) = 0 \qquad \Longrightarrow \qquad G = \bigl(G_0^{-1}+\Sigma\bigr)^{-1}, \tag{10}$$

$$\frac{\delta S}{\delta G(1,2)} = \Sigma(2,1) + \frac{\delta S_{\rm int}}{\delta G(1,2)} = 0 \qquad \Longrightarrow \qquad \Sigma = -\,\frac{\delta S_{\rm int}}{\delta G}. \tag{11}$$

Equation (10) is the Dyson equation $G^{-1} = G_0^{-1}+\Sigma$, and (11) identifies Σ as the self-energy generated by the interaction. For the free theory $S_{\rm int}=0$, so

$$\Sigma_\star = 0, \qquad G_\star = G_0, \tag{12}$$

the free propagator (4): the saddle of (9) exactly reproduces the microscopic two-point function, as it must.

Remark. The Dyson equation appears here as $G^{-1} = G_0^{-1}+\Sigma$ rather than the textbook $G^{-1} = G_0^{-1}-\Sigma$; the two differ only by the naming convention $\Sigma_{\rm here} = -\Sigma_{\rm textbook}$, fixed by the sign of the multiplier term in (6). Likewise the sign of $\operatorname{Tr}(\Sigma G)$ is correlated with the minus in the bilocal definition (5). The physical content — the Dyson equation and the saddle (12) — is convention-independent.

1.6 What the fluctuations are, and the bridge to §2

The free saddle (12) is trivial; the content is in the *fluctuations* of G about G_0, which (9) controls exactly. Writing $G = G_0+\delta G$ and eliminating Σ through (10), the Gaussian fluctuations of δG are the particle–hole pairs of the Fermi sea; their propagator is the free density–density response, and resumming them is the bilocal route to the random-phase approximation.

The nonperturbative reorganization used in §2 rests on a structural fact about the saddle. At zero temperature and equal times the free propagator reduces to the occupation $n_{\boldsymbol k}\in\{0,1\}$, so as a single-particle operator $\hat n$ it is a *projector*, $\hat n^2 = \hat n$. The configurations of G that the dynamics explores at low energy are not arbitrary: they are the images of $\hat n$ under canonical transformations of single-particle phase space (area-preserving deformations of the occupied region). That set is an orbit — the coadjoint orbit of the group of canonical transformations through the Fermi-sea projector — and restricting the G–Σ action to it, then expanding $-\operatorname{Tr}\ln$ in gradients, produces the Berry-phase (Kirillov–Kostant–Souriau) and Hamiltonian terms of the bosonized action. This is the Fermi-liquid analogue of the G–$\Sigma\to$ Schwarzian reduction in SYK, with the coadjoint orbit of canonical transformations in place of $\mathrm{Diff}(S^1)$. Section 2 carries this out.

2 From the free-fermion action to the coadjoint-orbit bosonization action

This section transcribes the derivation in the Appendix of Ye and Wang, *Phys. Rev. B* **112**, 035113 (2025), specialized to zero magnetic field. The aim is the same as §1 — recast the free-fermion theory in collective bosonic variables — but the route differs. Section 1 used the Grassmann coherent state and produced the bilocal G–Σ action. Here we use the *coherent state of the Fermi surface*: a group coherent state built from particle–hole deformations of the filled sea, the many-body analogue of a spin coherent state. The result is a manifestly geometric action whose target manifold is the coadjoint orbit of canonical transformations. Connecting this construction to the G–Σ action of §1 is the project's task; this document establishes the two endpoints independently.

2.1 Phase space, star product, and Moyal bracket

Single-particle observables of a d-dimensional system are functions $F(\boldsymbol{x}, \boldsymbol{p})$ on phase space (the paper works in $d = 2$). The reason phase space is the right arena is that the collective variable will be the occupied region of $(\boldsymbol{x}, \boldsymbol{p})$, and its deformations are canonical transformations. Operator multiplication maps not to pointwise multiplication of phase-space symbols but to the *star product*

$$F \star G = F \, \exp\Big(\tfrac{i\hbar}{2}\big(\overleftarrow{\partial}_{\boldsymbol{x}} \overrightarrow{\partial}_{\boldsymbol{p}} - \overleftarrow{\partial}_{\boldsymbol{p}} \overrightarrow{\partial}_{\boldsymbol{x}}\big)\Big) \, G, \tag{13}$$

and operator commutators map to the *Moyal bracket*

$$\{F, G\}_{\mathrm{M}} = -i\big(F \star G - G \star F\big) = 2F \, \sin\Big(\tfrac{\hbar}{2}\big(\overleftarrow{\partial}_{\boldsymbol{x}} \overrightarrow{\partial}_{\boldsymbol{p}} - \overleftarrow{\partial}_{\boldsymbol{p}} \overrightarrow{\partial}_{\boldsymbol{x}}\big)\Big) G. \tag{14}$$

Functions F equipped with $\{\cdot, \cdot\}_{\mathrm{M}}$ form the Moyal algebra $\mathfrak{g}$, the Lie algebra of canonical transformations; distribution functions f live in the dual space $\mathfrak{g}^*$, paired with observables through

$$\langle f, F \rangle \coloneqq \int \frac{\mathrm{d}^d x \, \mathrm{d}^d p}{(2\pi)^d} \, f(\boldsymbol{x}, \boldsymbol{p}) \, F(\boldsymbol{x}, \boldsymbol{p}). \tag{15}$$

In the semiclassical regime $\hbar \, \boldsymbol{\nabla}_{\boldsymbol{x}} \cdot \boldsymbol{\nabla}_{\boldsymbol{p}} \ll 1$ the star product reduces to ordinary multiplication and the Moyal bracket to the Poisson bracket $\{F, G\} = \partial_{\boldsymbol{x}} F \cdot \partial_{\boldsymbol{p}} G - \partial_{\boldsymbol{p}} F \cdot \partial_{\boldsymbol{x}} G$. Keeping the full Moyal algebra (not its Poisson truncation) is what later distinguishes the quantum theory; we set $\hbar = 1$ from here on except where shown.

2.2 The coadjoint orbit and the ground-state distribution

Canonical transformations form a group $\mathcal{G}$ whose Lie algebra is the Moyal algebra. Generated by $U = \exp(i\phi)$ with $\phi \in \mathfrak{g}$, they act on a distribution by conjugation,

$$f(\boldsymbol{x}, \boldsymbol{p}) = U \star f_0 \star U^{-1} \;\equiv\; \mathrm{Ad}^*_U \, f_0, \tag{16}$$

the coadjoint action on $\mathfrak{g}^*$. The base point is the ground-state distribution — the filled Fermi sea,

$$f_0(\boldsymbol{x}, \boldsymbol{p}) = \Theta\big(p_F - |\boldsymbol{p}|\big), \tag{17}$$

which is the phase-space (Wigner) image of the equal-time Fermi-sea projector $\hat{n}$ of §1.6. The set of all distributions reachable from f_0, $\mathcal{O}_{f_0} = \{\mathrm{Ad}^*_U f_0 : U \in \mathcal{G}\}$, is the *coadjoint orbit*; it is the target manifold of the effective theory. The physical content of (16) is that low-energy configurations are exactly the area-preserving deformations of the occupied region — no more, no less.

2.3 The target action

The bosonized action of the noninteracting Fermi surface is the sum of a Berry-phase term and a Hamiltonian term,

$$S[f] = S_{\mathrm{WZW}}[f] + S_H[f], \qquad S_{\mathrm{WZW}} = \int \mathrm{d}t\, \langle f_0,\, U^{-1} \star i\partial_t U \rangle, \qquad S_H = -\int \mathrm{d}t\, \langle f,\, \epsilon(\boldsymbol{p}) \rangle, \tag{18}$$

with $\epsilon(\boldsymbol{p}) = \varepsilon_0(\boldsymbol{p}) - \mu$ the dispersion measured from the chemical potential (the $\xi_{\boldsymbol{k}}$ of §1.1). The WZW term is the path integral of the Kirillov–Kostant–Souriau symplectic form on the orbit — the Berry phase accumulated as the occupied region deforms in time — and S_H is the single-particle energy. Its equation of motion $\delta S/\delta f = 0$ is the collisionless quantum Boltzmann equation $\partial_t f = \{\epsilon, f\}_{\mathrm{M}}$. The remainder of this section derives (18) from the microscopic theory.

2.4 Coherent state of the Fermi surface and its path integral

Let $|\mathrm{FS}\rangle$ denote the filled Fermi sea. A coherent state is obtained by dressing it with particle–hole deformations generated by fermion bilinears,

$$|\mathcal{U}\rangle = \hat{\mathcal{U}}\, |\mathrm{FS}\rangle, \qquad \hat{\mathcal{U}} = \exp\bigl(i\phi_{ij}\, c_i^\dagger c_j\bigr) \quad (i, j \in [1, L^d]), \tag{19}$$

the many-body analogue of a spin coherent state. Via the algebra of bilinears the $\hat{\mathcal{U}}$ form a Lie group, and free-fermion evolution $e^{i\hat{H}\,\mathrm{d}t}$ commutes with the group average; Schur's lemma then gives a resolution of identity over the Haar measure $\mathrm{d}\mathcal{U}$ (up to a constant),

$$\int \mathrm{d}\mathcal{U}\, |\mathcal{U}\rangle\langle\mathcal{U}| = 1. \tag{20}$$

Inserting (20) at every infinitesimal time slice — exactly as for the spin path integral — yields

$$Z = \int \mathcal{D}\mathcal{U}(t)\, \exp\Bigl[\int \mathrm{d}t\, \langle \mathrm{FS}|\, \hat{\mathcal{U}}^{-1}\bigl(-\partial_t - i\hat{H}\bigr)\hat{\mathcal{U}}\, |\mathrm{FS}\rangle\Bigr]. \tag{21}$$

2.5 Reduction to first-quantized operators

The exponent of (21) is a sum of nested commutators of bilinears. These close on themselves: for $\hat{A} = c_i^\dagger a_{ij} c_j$ and $\hat{B} = c_i^\dagger b_{ij} c_j$, fermion statistics give

$$[\hat{A}, \hat{B}] = c_i^\dagger \bigl(a_{ik} b_{kj} - b_{ik} a_{kj}\bigr) c_j = c_i^\dagger \bigl([\hat{a}, \hat{b}]\bigr)_{ij} c_j, \qquad \langle i|\hat{a}|j\rangle \equiv a_{ij}, \tag{22}$$

so the commutator of two bilinears is the bilinear of the *first-quantized* commutator. The many-body problem in the 2^N-dimensional Fock space therefore collapses onto $N \times N$ first-quantized operators. Re-exponentiating the nested commutators,

$$iS = \int \mathrm{d}t\, \langle \mathrm{FS}| c_i^\dagger c_j |\mathrm{FS}\rangle\, \bigl[\hat{U}^{-1}\bigl(-\partial_t - i\hat{\epsilon}\bigr)\hat{U}\bigr]_{ij}, \qquad \hat{U} \equiv e^{i\hat{\phi}}, \tag{23}$$

with $\hat\epsilon$ the one-particle Hamiltonian ($\hat H = c_i^\dagger \epsilon_{ij} c_j$). Introduce the one-particle density matrix (statistical matrix) of the ground state,

$$f_{0,ji} = \langle \mathrm{FS}|c_i^\dagger c_j|\mathrm{FS}\rangle, \tag{24}$$

so that $E_0 = \langle \mathrm{FS}|\hat H|\mathrm{FS}\rangle = \mathrm{Tr}(\hat f_0 \hat\epsilon)$. The action becomes a single first-quantized trace,

$$S = \int \mathrm{d}t\ \mathrm{Tr}\big[\hat f_0\, \hat U^{-1}\big(i\partial_t - \hat\epsilon\big)\hat U\big] = \int \mathrm{d}t\ \mathrm{Tr}\big[\hat f_0\, \hat U^{-1} i\partial_t \hat U - \hat f\, \hat\epsilon\big], \qquad \hat f \equiv \hat U \hat f_0 \hat U^{-1}, \tag{25}$$

where the second form uses cyclicity, $\mathrm{Tr}[\hat f_0 \hat U^{-1}\hat\epsilon\hat U] = \mathrm{Tr}[\hat f\,\hat\epsilon]$, and $\hat f$ (elements $\langle\Psi|c_i^\dagger c_j|\Psi\rangle$) is the one-particle density matrix of the coherent state.

2.6 Wigner transform to the phase-space action

Map first-quantized operators to phase-space symbols through the Wigner transform

$$\omega(\boldsymbol{x},\boldsymbol{p}) = \int \mathrm{d}^d y\; e^{i\boldsymbol{p}\cdot\boldsymbol{y}/\hbar}\,\big\langle \boldsymbol{x} - \tfrac{\boldsymbol{y}}{2}\big|\,\hat\omega\,\big|\boldsymbol{x} + \tfrac{\boldsymbol{y}}{2}\big\rangle. \tag{26}$$

Two properties close the derivation. A trace becomes a phase-space integral,

$$\mathrm{Tr}(\hat a\hat b) = \int \frac{\mathrm{d}^d x\, \mathrm{d}^d p}{(2\pi\hbar)^d}\, a(\boldsymbol{x},\boldsymbol{p})\, b(\boldsymbol{x},\boldsymbol{p}), \tag{27}$$

and the symbol of a product is the star product of symbols (so the symbol of a commutator is i times the Moyal bracket, (14)). Applying these to (25),

$$\boxed{S = \int \mathrm{d}t \int \frac{\mathrm{d}^d x\, \mathrm{d}^d p}{(2\pi\hbar)^d}\,\Big[\, f_0(\boldsymbol{x},\boldsymbol{p})\, U^{-1}\!\star i\partial_t U \;-\; f(\boldsymbol{x},\boldsymbol{p})\,\epsilon(\boldsymbol{p})\Big], \qquad f = U\star f_0 \star U^{-1}} \tag{28}$$

where $U(\boldsymbol{x},\boldsymbol{p}) = \exp[i\phi(\boldsymbol{x},\boldsymbol{p})]$ is the Wigner symbol of $\hat U$. The first term is $\langle f_0,\, U^{-1}\!\star i\partial_t U\rangle = S_{\mathrm{WZW}}$ and the second is $-\langle f,\epsilon\rangle = S_H$, so (28) is exactly the target action (18), with f on the coadjoint orbit (16). This completes the derivation of the bosonized action from the free-fermion theory.

Remark. The two sections meet at the ground state. The first-quantized density matrix $\hat f_0$ of (24) is the equal-time Fermi-sea projector $\hat n$ of §1.6; its Wigner symbol is $f_0(\boldsymbol{x},\boldsymbol{p}) = \Theta(p_F - |\boldsymbol{p}|)$, and the dispersion $\epsilon(\boldsymbol{p})$ here is the $\xi_{\boldsymbol{k}}$ of §1.1. The G–Σ fluctuations of §1.6 and the orbit coordinate $f = U\star f_0\star U^{-1}$ describe the same particle–hole physics from the two coherent-state representations; exhibiting their equality is the bridge the project constructs.

Caveats

- **Convention dependence.** The signs of Σ and of the $\mathrm{Tr}(\Sigma G)$ term in (9), and the explicit minus in G_0^{-1} (3), are tied to the choices (5) and (6). Different references distribute these signs differently; only the Dyson equation and the saddle (12) are invariant. Any comparison with §2 or with the literature must first align these conventions.

- **Regularization of** $\operatorname{Tr}\ln$**.** The functional determinant is UV-divergent; only the difference $\operatorname{Tr}\ln(G_0^{-1}+\Sigma)-\operatorname{Tr}\ln G_0^{-1}$ is finite, and equal-time objects such as the density require a point-splitting or frequency-sum prescription (the $e^{i\omega_n 0^+}$ convergence factor). These are left implicit above.

- **Measure and contour.** The bosonic measure $\mathcal{D}G\,\mathcal{D}\Sigma$ and the imaginary contour for Σ in (6) are stated formally; a careful treatment (e.g. of the Jacobian from $\psi,\bar{\psi}$ to G) is not needed for the saddle structure but matters for fluctuation determinants.

- **The two sections use different coherent states.** §1 is built on the Grassmann coherent state (yielding the bilocal G–Σ action); §2 on the group coherent state of the Fermi surface (yielding the coadjoint-orbit action). Both are exact rewritings of the same free-fermion theory, but their equality is *not* demonstrated here — it is the bridge the project constructs.

- **Real versus imaginary time.** §1 is formulated in imaginary time (Matsubara), §2 in real time t, following the respective sources. Aligning the two — including the $i\partial_t$ of the WZW term against the ∂_τ of the $\operatorname{Tr}\ln$ — is part of the reconciliation, not assumed here.

- **Moyal versus Poisson.** Section 2 keeps the full Moyal algebra; the Poisson-bracket form of the coadjoint action (e.g. in Delacrétaz–Else–Senthil) is its semiclassical truncation $\hbar\,\boldsymbol{\nabla}_{\boldsymbol{x}}\cdot\boldsymbol{\nabla}_{\boldsymbol{p}}\ll 1$. The distinction is physical, not cosmetic: it is what allows the magnetic extension of the source paper to capture Landau-level degeneracy.

- **Normalization and total-derivative subtleties of the WZW term.** The overall normalization of S_{WZW}, and a term linear in ϕ that is a total derivative in the infinite plane but becomes a topological θ-term once the momentum-space directions are compact, are treated in the body of the source paper (the weak-field theory), not in the zero-field transcription here.

- **Regularization of the traces.** Both the $\operatorname{Tr}\ln$ of §1 and the first-quantized Tr of §2 are over infinite single-particle Hilbert spaces and require the usual normal-ordering / point-splitting prescriptions for equal-time objects; these are left implicit.